Hugo Fragata

contact@hugofragata.com · github.com/hugofragata · Porto, Portugal

SUMMARY

Data Engineer with 8+ years of experience building and maintaining data platforms, pipelines, and near real-time data models. Comfortable owning the full data lifecycle on small, high-autonomy teams — from ingestion and transformation to observability, reporting, and enabling agentic access to data. AWS Certified Data Engineer – Associate. Native Portuguese speaker and fluent in English.

EXPERIENCE

Data Engineer — Web Summit ***Nov 2021 – Present***

- Own the data platform end-to-end on a two-person team, writing tickets, fielding internal requests, and driving annual KPIs
- Build and maintain near real-time data models and pipelines in Python, dbt, Polars, and SQL — ingesting from external APIs, Google Sheets, and PostgreSQL, orchestrated by Argo Workflows on Kubernetes via Docker images, stored in AWS S3, and served through Trino and Metabase, with GitHub powering versioning/PRs and CI/CD, and Hashicorp Vault handling secrets
- Single-handedly delivered 2 of 3 team's KPIs:
  - built a data staleness observability framework (incident definitions and alerts, response-time SLAs measurements, stakeholder-facing dashboard)
  - enabled agentic data access by creating a custom AI agent skill integrated with 1Password and Trino for natural-language querying of the data lake and models which saw high rates of adoption across many teams such as marketing, finance, engineering, product, growth
- Refactored the data compression pipeline module, replacing legacy Pandas/PyArrow code with Polars and consolidating a separate validation step into the fetch stage; cut internal compression time 25× (from ~40 minutes to ~90 seconds) while preserving Parquet-backed storage (10× smaller) and query performance (up to 13× faster)

Data Infrastructure Engineer — DefinedCrowd (Defined.ai) ***Mar 2020 – Nov 2021***

- Built data infrastructure for an AI training data platform spanning 70+ languages serving multiple Fortune 500 clients

Data Engineer — BMW Group (via Critical TechWorks) ***Sep 2019 – Mar 2020***

- Developed data ingestion pipelines from BMW Group sources into the on-premises data lake

Data Engineer — Vodafone Spain (via Altran Consulting) ***Aug 2018 – Aug 2019***

- Built high-performance big data clusters from bare-metal provisioning through ingestion, storage, and analytics extraction at speed

CERTIFICATIONS

- AWS Certified Data Engineer – Associate — Amazon Web Services *Nov 2026*

EDUCATION

MSc Computer and Telematics Engineering — Universidade de Aveiro ***2013 – 2018***
Completed all coursework and credits; thesis pending. Published two papers on predictive machine learning models for banking (RECPAD 2017, EPIA 2017).

Astronomy and Astrophysics — Universidade do Porto ***2012 – 2013***